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AWS Storage Services

Secure, scalable and reliable storage for every workload

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PEOPLE SKILLS OPPORTUNITIES

AWS provides a wide range of storage services designed for different use cases, from object storage to block storage, file storage and backup. Choose the right service based on your data type, access pattern, performance and cost requirements.

Amazon S3

Object Storage



Scalable object storage for any amount of data. Ideal for backups, data lakes, static websites, media files and more.

Key Features

- Highly durable (99.999999999%)
- Scalable and cost-effective
- Multiple storage classes (Standard, IA, Glacier, etc.)
- Versioning and lifecycle management
- Event notifications

Common Use Cases

- Backup and archiving
- Static website hosting
- Data lakes and analytics
- Media storage (images, videos)
- Log storage

Amazon EBS

Block Storage



Persistent block storage volumes for EC2 instances. Provides high performance and low latency.

Key Features

- High performance and low latency
- Multiple volume types (gp3, io2, st1, sc1)
- Snapshots for backup
- Customizable size and performance
- Encrypted by default (optional)

Common Use Cases

- Boot volumes for EC2
- Databases
- Application storage
- Transactional workloads

Amazon EFS

File Storage



Managed, scalable file storage for Linux instances. Accessible by multiple EC2 instances simultaneously.

Key Features

- Scales automatically
- Pay for what you use
- Supports NFS
- High availability across Availability Zones
- Encrypted at rest and in transit

Common Use Cases

- Shared file storage
- Content management systems
- Development environments
- Big data and analytics
- Home directories

Amazon FSx

Managed File Systems



Fully managed file systems for specific workloads (Windows, Lustre, NetApp ONTAP and OpenZFS).

Key Features

- Multiple file system options
- High performance
- Fully managed by AWS
- Integrates with Active Directory (for Windows)
- Built for enterprise workloads

Common Use Cases

- Windows file shares (FSx for Windows File Server)
- High performance computing (FSx for Lustre)
- Enterprise applications (FSx for NetApp ONTAP)
- Modern Linux workloads (FSx for OpenZFS)

Amazon Glacier

Archive Storage



Low-cost storage for long-term archiving and backup with multiple retrieval options.

Key Features

- Very low cost
- Designed for long-term storage
- Multiple retrieval options (Instant, Standard, Bulk)
- Integrated with S3 lifecycle policies
- Highly durable

Common Use Cases

- Long-term data archiving
- Compliance and regulatory retention
- Disaster recovery
- Cold data storage

AWS Storage Gateway

Hybrid Storage



Connects your on-premises environment to AWS storage services for a hybrid cloud solution.

Key Features

- Integrates with S3, EBS and FSx
- Supports file, volume and tape gateways
- Seamless hybrid storage
- Cached and tiered storage
- Easy to deploy and manage

Common Use Cases

- Hybrid cloud storage
- Data migration
- Backup and disaster recovery
- Extending on-premises storage to AWS

How the Services Fit Together



Choosing the Right Storage Service

Requirement / Use Case	Recommended Service
Store any type of unstructured data	**Amazon S3**
Persistent storage for EC2	**Amazon EBS**
Shared file storage for Linux	**Amazon EFS**
Managed file systems (Windows/Lustre/NetApp)	**Amazon FSx**
Long-term, low-cost archiving	**Amazon Glacier**
Hybrid cloud storage (on-premises to AWS)	**AWS Storage Gateway**

Key Takeaways

- Use S3 for scalable and cost-effective object storage.
- Use EBS for high-performance block storage with EC2.
- Use EFS for shared file storage across multiple instances.
- Use FSx for specialized file system workloads.
- Use Glacier for long-term, low-cost archiving.
- Use Storage Gateway to connect on-premises storage to AWS.
- Choose based on your data type, access pattern, performance and cost requirements.

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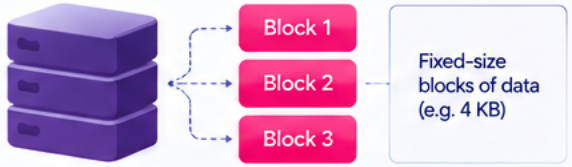
Block vs Object vs File Storage

Different storage types for different needs

AWS provides multiple storage options because different workloads have different requirements. Understanding the differences helps you choose the right storage for your applications.

1 Block Storage

Raw storage volumes for high-performance workloads



Example: Amazon EBS

Block storage divides data into fixed-size blocks. Each block is addressed individually and appears as a disk to your operating system. It is designed for high performance and low latency.

Common Use Cases

- Databases (e.g. MySQL, PostgreSQL, SQL Server)
- Application servers
- High-performance computing (HPC)
- Boot volumes for EC2 instances

AWS Services

- Amazon EBS (Elastic Block Store)
- Instance Store (ephemeral block storage)

Advantages

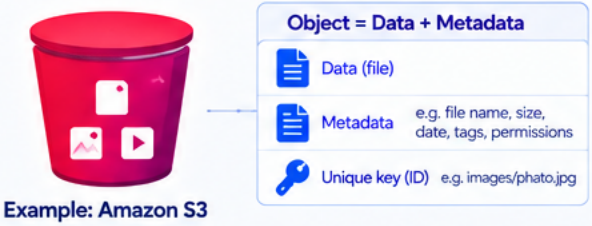
- High performance and low latency
- Ideal for transactional workloads
- Can be easily attached to EC2 instances

Considerations

- Only accessible from one instance (at a time)
- Not ideal for large, unstructured data
- You manage scaling (e.g. increase volume size)

2 Object Storage

Scalable storage for unstructured data



Example: Amazon S3

Object storage stores data as objects (files) along with metadata and a unique identifier. It is designed for massive scale, high durability and global accessibility over the internet.

Common Use Cases

- Storing backups and archives
- Media files (images, videos, audio)
- Data lakes and big data analytics
- Static websites
- Application data and logs

AWS Services

- Amazon S3 (Simple Storage Service)
- S3 Glacier (for long-term archiving)

Advantages

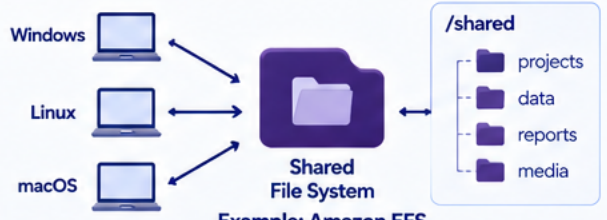
- Virtually unlimited storage capacity
- Highly durable (designed for 99.999999999% durability)
- Accessible from anywhere via API/HTTP
- Cost-effective
- No infrastructure to manage

Considerations

- Not designed for low-latency, transactional workloads
- Not mounted as a file system (uses APIs instead)
- Not suitable for applications that require frequent data modification in place

3 File Storage

Shared file systems for multiple users and applications



Example: Amazon EFS

File storage provides a shared file system that can be accessed by multiple users and instances at the same time. It uses standard file protocols such as NFS (and SMB for Windows).

Common Use Cases

- Shared storage for web servers and application servers
- Content management systems
- Development and test environments
- Media processing and rendering
- Home directories

AWS Services

- Amazon EFS (Elastic File System) – NFS
- Amazon FSx (for Windows File Server, Lustre, NetApp ONTAP, etc.)

Advantages

- Shared access from multiple instances
- Simple to use (standard file system interface)
- Scales automatically
- Ideal for collaborative workloads

Considerations

- Higher cost than object storage
- Generally lower performance than block storage
- Not suitable for transactional workloads (e.g. databases)

Key Takeaway

There is no single best storage type. Choose block storage for high performance, object storage for massive scale and unstructured data, and file storage for shared access.

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Amazon EBS (Elastic Block Store)

High-performance block storage for your EC2 instances

Amazon EBS provides persistent, high-performance block storage volumes that you can attach to EC2 instances.
It is designed for high availability, durability and flexibility, making it ideal for databases, applications and general-purpose workloads.



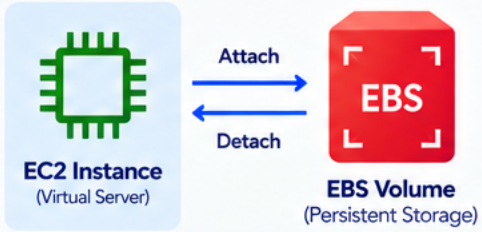
What is Amazon EBS?



Amazon Elastic Block Store (EBS) is a scalable block storage service that provides persistent storage for EC2 instances. EBS volumes behave like physical hard drives and remain available even if you stop or reboot your instance.

- **Persistent**
  Data remains after instance stop or reboot
- **High performance**
  Optimized for different workloads
- **Highly durable**
  Data is replicated within an Availability Zone
- **Scalable**
  Easily increase or decrease size as needed

How Amazon EBS Works



- Data persists even if the instance is stopped
- Can be resized on the fly
- Snapshots can be taken for backup

Key Use Cases

- **Databases**
  MySQL, PostgreSQL, SQL Server, etc.
- **Application servers**
  Web and enterprise applications
- **General purpose storage**
  File systems and data storage
- **Development and testing**
  Flexible and cost-effective storage
- **Backup and recovery**
  Use snapshots for backup and disaster recovery

EBS Volume Types

Amazon EBS offers different volume types to suit various workloads.

gp3
General Purpose

- Balanced performance and cost
- Suitable for most workloads
- Baseline performance with ability to scale

io2
Provisioned IOPS

- High performance
- Low latency
- For critical databases and business applications

st1
Throughput Optimized

- Low cost
- High throughput
- Ideal for big data, analytics and frequently accessed data

sc1
Cold HDD

- Lowest cost
- Designed for infrequently accessed data
- Ideal for archiving and backup

EBS Snapshots

Snapshots are point-in-time backups of your EBS volumes stored in Amazon S3.



Snapshot Benefits

- Durable and stored in Amazon S3
- Can be used to create new volumes
- Helps with backup, disaster recovery and migration
- Incremental – only changes are stored

Key Takeaways

- Amazon EBS provides persistent block storage for EC2 instances.
- Multiple volume types (gp3, io2, st1, sc1) for different workloads.
- Data remains available even if your instance is stopped.
- Snapshots enable easy backup and recovery.
- You can resize volumes and change volume types as needed.

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Amazon S3 (Simple Storage Service)

Scalable, durable and secure object storage for any amount of data

Amazon S3 is an object storage service that allows you to store and retrieve any amount of data from anywhere on the internet.
It is designed for high durability, high availability and scalability, making it ideal for backups, data lakes, static websites, media files and more.



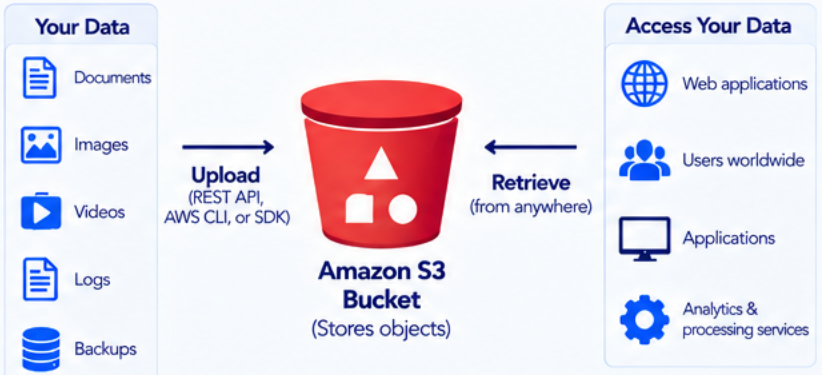
What is Amazon S3?



Amazon S3 (Simple Storage Service) is a highly scalable object storage service that stores data as objects in buckets. You can store and retrieve any amount of data, at any time, from anywhere on the internet.

- Store any type of data (files, images, videos, logs, etc.)
- Virtually unlimited storage capacity
- High durability (designed for 99.999999999% durability)
- Accessible from anywhere via the internet
- Pay only for what you use

How Amazon S3 Works



Key Features

High Durability
Designed for 99.999999999% durability.

Highly Scalable
Automatically scales to store any amount of data.

Secure
Built-in security features like encryption, access control and monitoring.

Cost-effective
Pay only for the storage you use.

Global Access
Access your data from anywhere on the internet.

Common Use Cases

- Backup and disaster recovery
- Data lakes and big data analytics
- Static website hosting
- Media storage (images, videos, audio)
- Application data and logs
- Software distribution and data sharing

S3 Storage Classes

S3 Standard	Frequently accessed data
S3 Intelligent-Tiering	Automatically moves data to lower cost tiers
S3 Standard-IA	Infrequently accessed data
S3 One Zone-IA	Infrequently accessed data (ONE Availability Zone)
S3 Glacier Instant Retrieval	Long-term archive (milliseconds)
S3 Glacier Flexible Retrieval	Long-term archive (minutes to hours)
S3 Glacier Deep Archive	Lowest cost archive (hours)

S3 Bucket and Object



Key Takeaways

- Amazon S3 stores data as objects in buckets.
- It is highly durable, scalable and secure.
- Choose the right storage class for your use case.
- You can access your data from anywhere using the internet.
- S3 is widely used for backups, data lakes, web hosting and media storage.

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S3 Bucket Policies

Control who can access your S3 buckets and what they can do

A bucket policy is a JSON-based access policy that you attach to an Amazon S3 bucket. It defines who (principals) can perform which actions on which resources in the bucket, and under what conditions.

What is a Bucket Policy?



A bucket policy is a resource-based policy that is attached directly to an S3 bucket. It controls access to the bucket and the objects in it.

- Written in JSON format
- Attached to a specific bucket
- Grants or denies access
- Supports fine-grained permissions
- Can be used with IAM users, roles, other AWS accounts or public access

Bucket Policy Structure

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Sid": "AllowGetObject",
      "Effect": "Allow",
      "Principal": "*",
      "Action": "s3:GetObject",
      "Resource": "arn:aws:s3:::example-bucket/*",
      "Condition": {
        "StringEquals": {
          "aws:Referer": "https://www.example.com"
        }
      }
    }
  ]
}
```

1. **Version** — The policy language version (e.g. 2012-10-17).
2. **Statement** — A list of one or more statements.
3. **Effect** — Allow or Deny.
4. **Principal** — Who the policy applies to (e.g. AWS account, IAM user, role, or * for everyone).
5. **Action** — The actions allowed or denied (e.g. s3:GetObject, s3:PutObject).
6. **Resource** — The bucket or object this policy applies to.
7. **Condition** (optional) — Additional conditions (e.g. IP address, source VPC, referer, encryption, etc.).

Common Use Cases

- **Public read access** — Allow anyone to view objects (e.g. for a static website).
- **Restrict access to specific users or accounts** — Allow only your AWS account or a specific IAM role to access the bucket.
- **Allow access from specific IP addresses** — Restrict access to a trusted IP range.
- **Enforce secure access (HTTPS)** — Allow access only if requests use HTTPS.
- **Allow access from a specific VPC** — Restrict access to requests from your VPC.
- **Grant access to specific actions** — Allow only certain actions (e.g. GetObject) on specific objects or prefixes.

Example: Public Read Access

Allow anyone to read objects in a bucket.

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Effect": "Allow",
      "Principal": "*",
      "Action": "s3:GetObject",
      "Resource": "arn:aws:s3:::example-bucket/*"
    }
  ]
}
```

Example: Restrict to Specific IP

Allow access only from a specific IP address range.

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Effect": "Allow",
      "Principal": "*",
      "Action": "s3:GetObject",
      "Resource": "arn:aws:s3:::example-bucket/*",
      "Condition": {
        "IpAddress": {
          "aws:SourceIp": "203.0.113.0/24"
        }
      }
    }
  ]
}
```

Example: Enforce HTTPS

Deny any request that is not using HTTPS.

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Sid": "DenyUnsecureTransport",
      "Effect": "Deny",
      "Principal": "*",
      "Action": "s3:*",
      "Resource": "arn:aws:s3:::example-bucket/*",
      "Condition": {
        "Bool": {
          "aws:SecureTransport": "false"
        }
      }
    }
  ]
}
```

Best Practices

- Follow the principle of least privilege.
- Use specific principals and conditions.
- Enforce HTTPS for secure access.
- Use AWS IAM roles where possible.
- Test your policies before applying to production.
- Regularly review and update your policies.

Key Takeaways

- Bucket policiee give you fine-grained control over who can access your S3 data.
- They are written in JSON and attached to the bucket.
- Use conditions to improve security.
- Always follow security best practices.

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S3 Lifecycle Policies

Automatically manage your data to reduce costs

S3 Lifecycle policies help you automatically transition or expire objects in your bucket based on rules you define.
This allows you to store data in the most cost-effective storage class and remove data that is no longer needed.

How Lifecycle Policies Work

You define rules based on object age, and S3 automatically transitions or expires the objects.



Day 0
S3 Standard
(Frequently accessed)

S3 Standard-IA
(Infrequently accessed)

S3 Glacier
(Archive)

Delete
(Expired)

Lifecycle policies run automatically. You do not need to manually move or delete objects.

Lifecycle Policy Rule Structure

```
{
  "Rules": [
    {
      "ID": "Move to IA and Glacier",
      "Status": "Enabled",
      "Filter": {
        "Prefix": "logs/"
      },
      "Transitions": [
        {
          "Days": 30,
          "StorageClass": "STANDARD_IA"
        },
        {
          "Days": 90,
          "StorageClass": "GLACIER"
        }
      ],
      "Expiration": {
        "Days": 365
      }
    }
  ]
}
```

Common Lifecycle Actions

Transition to a lower-cost storage class
Move objects to S3 Standard-IA, Glacier Instant Retrieval, Glacier or Deep Archive.

Expire objects
Automatically delete objects after a specified number of days.

Transition current versions
Apply rules to the current version of objects.

Handle previous versions
Apply different rules to previous versions (if versioning is enabled).

Filter by prefix or tags
Apply rules to specific folders (prefixes) or objects with certain tags.

S3 Storage Classes for Lifecycle

S3 Standard
For frequently accessed data.

S3 Standard-IA
For infrequently accessed data (lower cost).

S3 Glacier Instant Retrieval
For rarely accessed data, milliseconds retrieval.

S3 Glacier
For long-term archive (minutes to hours retrieval).

S3 Glacier Deep Archive
For long-term archive (hours retrieval, lowest cost).

Example Use Case

Automatically manage application logs to reduce storage costs.



0 – 30 days
S3 Standard
(Fast access)

30 – 90 days
S3 Standard-IA
(Lower cost)

90 – 365 days
S3 Glacier
(Archive)

After 365 days
Delete
(Remove old data)

Use lifecycle policies for logs, backups, media files, and any data that becomes less important over time.

Benefits

- Reduce storage costs automatically
- No manual intervention required
- Keeps your data organized
- Supports compliance and data retention policies
- Works at scale for large amounts of data

Best Practices

- Use prefixes or tags to apply rules to specific data
- Review and test your rules before applying to production
- Consider your data access patterns and compliance needs
- Use S3 Inventory to analyze your data before creating rules
- Regularly review and update your lifecycle policies

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Amazon S3 Storage Classes

The right storage for the right data

Amazon S3 offers multiple storage classes, each designed for different access patterns, performance needs and costs. You can choose the best class based on how often you need to access your data and how long you need to keep it.

S3 Standard (General Purpose)

Best for frequently accessed data with low latency and high throughput.

- **Availability** 99.99%
- **Durability** 99.9999999999% (11 9s)
- **Access** Frequently
- **Storage duration** Short to medium
- **Cost** Highest (per GB)

S3 Intelligent-Tiering (Automatic)

Automatically moves data between access tiers based on your usage patterns.

- **Availability** 99.9%
- **Durability** 99.9999999999% (11 9s)
- **Access** Frequent to infrequent (automatic)
- **Storage duration** Long-term
- **Cost** Variable (based on access pattern)

S3 Standard-IA (Infrequent Access)

Lower cost than Standard for data that is accessed infrequently.

- **Availability** 99.9%
- **Durability** 99.9999999999% (11 9s)
- **Access** Infrequent
- **Storage duration** Long-term
- **Cost** Lower than Standard (per GB)

S3 One Zone-IA (Infrequent Access)

Lower cost storage for infrequently accessed data that can be kept in a single Availability Zone.

- **Availability** 99.5% (single AZ)
- **Durability** 99.9999999999% (11 9s)
- **Access** Infrequent
- **Storage duration** Long-term
- **Cost** Lower than Standard-IA (per GB)

S3 Glacier Instant Retrieval (Archive – Instant)

Low-cost, long-term storage with millisecond retrieval for data you need quickly (usually within seconds).

- **Availability** 99.9%
- **Durability** 99.9999999999% (11 9s)
- **Access** Instant (milliseconds)
- **Storage duration** Long-term
- **Cost** Lower than Standard-IA (per GB)

S3 Glacier Deep Archive (Archive – Bulk)

Lowest cost storage for data that you rarely access and can tolerate hours of retrieval time.

- **Availability** 99.9%
- **Durability** 99.9999999999% (11 9s)
- **Access** Hours (typically 12+)
- **Storage duration** Long-term
- **Cost** Lowest (per GB)

When to use each storage class

Standard
- Websites
- Mobile apps
- Big data analytics
- Real-time applications

Intelligent-Tiering
- Unpredictable access
- Data lakes
- User generated content
- Backup data

Standard-IA
- Backups
- Disaster recovery
- Business continuity
- Archival (medium term)

One Zone-IA
- Log files
- Data processing
- Secondary backups
- Non-critical data

Glacier Instant Retrieval
- Compliance data
- Healthcare records
- Research data
- Archive with quick access

Glacier Deep Archive
- Long-term backups
- Rarely accessed data
- Regulatory archives
- Digital preservation

Key Takeaways

- Choose a storage class based on access frequency, performance needs and cost.
- You can move objects between storage classes using lifecycle policies.
- Intelligent-Tiering automatically optimizes costs for you.
- Glacier Instant Retrieval and Deep Archive are ideal for long-term archival data.
- The right storage class helps you balance performance, durability and cost.

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Static Website Hosting with Amazon S3

Fast. Scalable. Secure. Cost-Effective.

Host your static websites (HTML, CSS, JavaScript, images, etc.) on Amazon S3 and deliver them to the world.
No servers to manage. Pay only for what you use.

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What is S3 Static Website Hosting?



Amazon S3 can host static websites by storing your website files (HTML, CSS, JavaScript, images, etc.) in a bucket and serving them over the internet.

- No servers to manage
- Highly scalable and durable
- Low cost (pay only for storage and data transfer)
- Ideal for personal sites, landing pages, documentation and web apps (frontend)

How It Works



Use Cases

- **Personal websites** — Portfolio, resumes, blogs
- **Landing pages** — Marketing campaigns, event pages
- **Documentation sites** — Project docs, open source projects
- **Frontend for web applications** — (Use with services like API Gateway, CloudFront, etc.)
- **Media and static content hosting** — Images, PDFs, downloads

Step-by-Step Guide

1. **Create an S3 bucket**
   Choose a unique bucket name (e.g., my-website-123).
2. **Upload your website files**
   Upload index.html, CSS, JavaScript, images, etc.
3. **Enable static website hosting**
   In the bucket properties, enable Static website hosting and set index document (e.g., index.html).
4. **Set permissions**
   Update the bucket policy to allow public read access (for a simple website).
5. **Access your website**
   Use the S3 website endpoint URL to view your site.

Bucket Policy Example (Public Read Access)

JSON

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Sid": "PublicReadGetObject",
      "Effect": "Allow",
      "Principal": "*",
      "Action": "s3:GetObject",
      "Resource": "arn:aws:s3:::my-website-123/*"
    }
  ]
}
```

Example Website URL

http://my-website-123.s3-website-region.amazonaws.com

Note: Replace my-website-123 with your bucket name and region with your AWS region (e.g., us-east-1).

Key Takeaways

- S3 static website hosting is simple and cost-effective.
- Use a unique bucket name and enable static website hosting.
- Set the correct permissions with a bucket policy.
- For custom domains and HTTPS, use Amazon CloudFront with Route 53.
- Ideal for static content (HTML, CSS, JavaScript, images, etc.).

Best Practices

- Use CloudFront for HTTPS and better performance.
- Use a custom domain with Route 53.
- Keep your content optimized (minified files, compressed images).
- Use versioning and lifecycle policies if needed.
- Restrict write access (only allow public read).

Limitations

- Only static content (no server-side code).
- Not suitable for dynamic applications (backend logic).
- No built-in HTTPS (use CloudFront for SSL/TLS).
- Bucket must be public (or use CloudFront with OAC for private buckets).

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Amazon S3 vs EBS vs EFS

Different storage. Different use cases. Choose the right one.

AWS provides multiple storage options, each designed for different needs. Understanding the differences helps you build secure, scalable and cost-effective solutions.

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Feature	Amazon S3 **Object Storage** Store and retrieve any amount of data from anywhere	Amazon EBS **Block Storage** High-performance storage for a single EC2 instance	Amazon EFS **File Storage** Shared, scalable file system for multiple EC2 instances
Storage type	Object storage (stores data as objects)	Block storage (raw block device)	File storage (NFS file system)
Access method	HTTPS (REST API, SDK, CLI, Console)	Attached to a single EC2 instance (as a drive)	Mounted via NFS (multiple EC2 instances)
Data structure	Objects (files) with metadata	Blocks (like a hard drive)	Files and directories (POSIX compliant)
Capacity	Virtually unlimited	Up to 64 TB per volume (gp3, io2, etc.)	Automatically scales to petabytes
Scalability	Highly scalable (no capacity planning)	Fixed size (can be increased)	Automatically scales (elastic)
Durability	99.999999999% (11 9s)	Designed for high durability (backed within an AZ)	High durability (multi-AZ)
Availability	Highly available (across multiple AZs)	Single AZ (tied to one EC2 instance)	Multi-AZ (highly available)
Performance	Good for large files, high throughput	Low-latency, high IOPS (optimized for EC2)	Good for shared file access, scales with usage
Typical use cases	Backups, archives, static websites, media files, data lakes, logs	Operating systems, databases, application data	Shared files, web content, development environments, content management
Access from multiple EC2 instances	Yes (via internet or VPC endpoints)	**No** (can only be attached to one instance at a time)	**Yes** (multiple instances simultaneously)
Cost	Low (pay per storage and requests)	Higher (pay per GB and provisioned performance)	Higher than S3 (pay per GB used)
Best for	Massive, unstructured data and global access	High-performance storage for a single EC2 instance	Shared, elastic file system for multiple EC2 instances

Example Use Cases

Amazon S3
- Host a static website
- Store backups
- Store application logs
- Store images, videos and documents
- Data lake for analytics

Amazon EBS
- Operating system volume
- Database storage (e.g., MySQL, PostgreSQL)
- Application data (e.g., web servers)
- High IOPS workloads
- Temporary or persistent block storage

Amazon EFS
- Shared web content (e.g., WordPress)
- Shared development environments
- Media processing workflows
- Home directories
- Applications that need shared file access

Key Takeaway Use S3 for massive, low-cost object storage. Use EBS for high-performance block storage on a single EC2 instance. Use EFS for a shared, scalable file system across multiple EC2 instances.